ENTERPRISE SOFTWARE TEST DEVELOPMENT ENGINEER (RDSS INTERN)

Software Development, Testing & Reliability

ABC Mapper / NeoPower System

Yu-Jui Liang
Data Science Master's Degree Program
National Taiwan University | 20 Oct 2026

Software Development / Verification & Testing / Debugging

Education & Professional Experience

01 / BACKGROUND

Education

National Taiwan University

M.S. Candidate, Data Science Master's Degree Program

Expected Graduation: June 2027

National Chengchi University

B.S. in Computer Science

Double Major in Electronic Physics

Sep. 2021 – Jun. 2025

Professional Experience

AIMobile Co., Ltd.

Inventec Group

Contract Software Engineer

Apr. 2025 – Jul. 2025

Neopower Technologies Co., Ltd.

Contract Software Engineer

Jun. 2023 – Mar. 2025

Outline

02 / PRESENTATION OVERVIEW

Two projects, two complementary software engineering experiences.

01 ABC Mapper
Software Extension & Integration

Modular C++ development, backward compatibility, and regression testing.

02 NeoPower
Backend Development & Maintenance

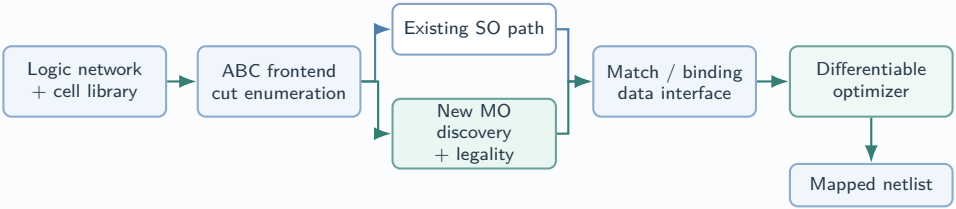
API integration, system operations, simulation, and debugging in a smart charging platform.

Focus: engineering decisions, verification strategies, and lessons from implementation.

ABC Mapper: Software Extension & Integration

Extend an existing EDA tool while preserving its original single-output flow.

Architecture: Add a Parallel MO Path



Non-invasive Software Extension

Independent MO candidate discovery and output-pair binding; legacy SO workflow kept separate.

Modular Toolchain Integration

ABC match export → differentiable SO/MO selection → netlist reconstruction.

ABC Mapper: Correctness & Regression Testing

New mapping features must preserve the original Boolean behavior.

01 Compatibility

SO-only smoke / regression cases verify that the pre-existing flow is not disrupted.

02 Candidate Legality

Check MO output roles, pin compatibility, pair structure, and match-data consistency.

03 End-to-End CEC

Compare original network behavior against the generated mapped netlist.

Implementation-to-Verification Loop

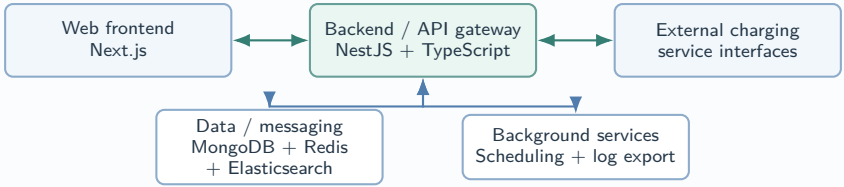


Validation principle: CEC proves functional equivalence; area / delay optimization is evaluated separately.

NeoPower: System Architecture & Integration

An electric vehicle (EV) charging management platform integrating APIs and backend services.

High-Level Service Architecture



Backend / API Development

Service boundaries, request-response contracts, and integrations with charging operations.

Container-Based Service Isolation

Isolated individual services in containers to simplify deployment, maintenance, and troubleshooting.

NeoPower: Testing & Reliability Engineering

Validate changes through real-world simulation, regression testing, and field evaluation.

Pre-Deployment Testing

- External vendor / client simulators
- Production-derived test scenarios
- High load and multi-charger failures
- Scenario-based regression testing

Post-Deployment A/B Testing

- Compare existing and updated implementations
- Time-based testing at the same charging site
- Cross-site comparison under similar conditions
- Operational metrics and customer feedback

Field-Informed Regression Workflow



Historical incidents become reusable test scenarios for future software releases.

Thank You

Questions & Discussion